Syllable token frequencies reveal a strong cross-linguistic prevalence of the Sonority Sequencing Principle

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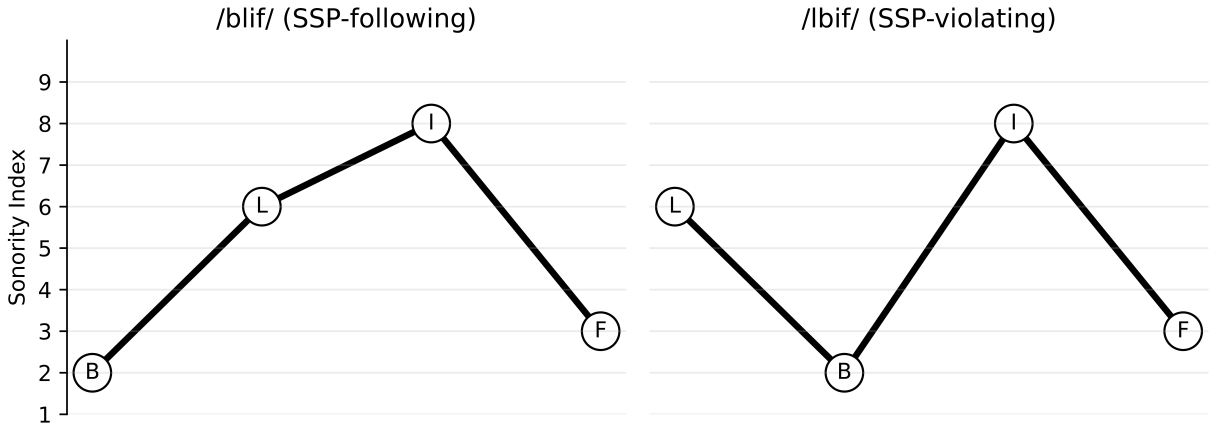
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Sonority is a phonological feature that describes the perceptual salience of a given phoneme relative to neighbouring ones (Blevins, 1995; S. Parker, 2008; S. G. Parker, 2002; Selkirk, 1984). While the acoustic correlates of sonority are still under debate, some physical features like intensity have been shown to strongly correlate with sonority indexes across spoken languages (Fant et al., 2000; Jany et al., 2007; S. Parker, 2017; S. G. Parker, 2002; but see Albert & Nicenboim, 2022 for a different perspective). Phonemes can be hierarchically organised according to their sonority, with those at the lower end having lower sonority and those at the higher end having higher sonority (e.g., Obstruents < Nasals < Liquids < Glides < Vowels, Clements, 1990). This hierarchy is often represented as a numerical scale (e.g., 1 for lowest sonority, 9 for highest sonority, S. Parker, 2008). Based on this sonority hierarchy, some syllables are considered well-formed across languages while others are marked as ill-formed. The Sonority Sequencing Principle (SSP) is a phonotactic rule that captures how sonority constrains the ordering of phonemes within a syllable. The SSP states that each syllable should display a unique peak of sonority at the nucleus, and that sonority should rise toward the nucleus and fall away from it (S. G. Parker, 2002; Selkirk, 1984). Syllables are classified as following or violating the SSP by examining the sequence of sonority values assigned to each of its phonemes. Figure 1 shows an example: the syllable /blif/ has a sonority contour of 2-6-8-3 (8 being the vowel /i/, nucleus of the syllable) and is therefore considered well-formed, whereas /lbif/ (6-2-8-3) is considered ill-formed because the onset displays a sonority dip from the first to the second phoneme, violating the progressive increase in sonority that characterizes well-formed syllables.

Figure 1

*Schematic representation of the sonority hierarchy. Examples of SSP-following (*blif*) and SSP-violating syllables (*lbif*) are provided according to a numerical sonority index scale (1-9).*



Experimental evidence suggests that the SSP plays a pivotal role in processing spoken (Berent et al., 2007, 2011; Berent & Lennertz, 2010; Miozzo & Buchwald, 2013; S. Parker, 2017) and sign languages (Perlmutter, 1993; Sandler, 1993; Stone et al., 2018). Studies on newborns, infants, and adults show a consistent preference toward well-formed syllables (SSP-following) over ill-formed ones (SSP-violating), even when well-formed syllables are unattested in the participants' native language. For example, English-native adults prefer /bn/ (SSP-following) over /lf/ (SSP-violating) at syllable onset, although neither consonant cluster occurs in English (Berent et al., 2011; Berent & Lennertz, 2010). Few-day-old newborns show different brain haemodynamic responses when listening to SSP-following or violating syllables (e.g., *bran* vs. *lbif*) suggesting that precursors of the SSP are already present at birth (García-Castro et al., 2026; Gomez et al., 2014). Such preferences for sonority-compliant syllables, even with unattested materials or very limited language experience, as in newborn infants, have

been considered supporting evidence for the status of the SSP as a linguistic universal, with which humans might be equipped from birth.

The universal status of the SSP is, however, still a matter of ongoing debate. Violations of the SSP are not uncommon across languages, with examples including fricative-stop consonant clusters at syllable onset (e.g., /ʃt/, as in the German word *Stein*) and their reverse counterpart at coda, stop-fricative clusters (e.g., /ks/, as in the English word *clicks*) (Kawasaki-Fukumori et al., 1997; Morelli et al., 2003). Languages that allow considerably more complex consonant clusters at syllable onset tend to show SSP violations more frequently. This for instance the case of Polish, which allows for clusters of up to three consonants at syllable onset (e.g., *chrząszcz* /ˈxʂɔ̃wʂt͡ʂ/ [beetle]) (Gussmann, 2007; Jarosz, 2017), or Georgian (e.g., მწვრთნელი /mt͡sʼvrtʰneli/ [trainer]) (Crouch et al., 2023).

To our knowledge, only two studies have provided a quantitative approach toward the cross-linguistic prevalence of the SSP. Evidence of a preference for SSP-following consonant clusters across 43 languages was reported, with type frequencies of 97 distinct consonant clusters reflecting an overall 72% compliance with the SSP (Frisch, 2015). When type frequencies were analyzed across 496 languages from 58 language families, it was found that 32% of the consonant clusters at word onsets violated the SSP, compared with 30% at word-final position (Yin et al., 2023). Both studies relied on type frequencies of attested consonant clusters in a sample of languages, counting the number of consonant clusters that followed or violated the SSP in each of those languages, or simply establishing the presence or absence of SSP violations in them. While this approach provides useful information about languages' tolerance of SSP violations (or lack thereof), it does not provide a direct estimate of how often speakers of such languages encounter SSP violations in natural speech. This is

because the prevalence of a given phonological structure in a language (type frequency) does not necessarily correspond on a one-to-one basis to its frequency in natural speech. Consider an imaginary language that allows for 20 consonant clusters in total, one of which is /st/. While /st/ represents a sizeable portion of the language's phonotactical repertoire (5%), how frequently speakers encounter /st/ depends on at least two factors: the number of words containing /st/, and the frequency of each of those words. If /st/ occurs in only a few words or only in low-frequency words, its share in naturalistic speech would be lower than suggested by the language's phonological repertoire.

Another pitfall is that the phonological transcriptions used in most studies are drawn from lexical inventories and often lack syllable boundaries (Yin et al., 2023), making it difficult to draw reliable conclusions about the prevalence of the SSP in natural speech. The lack of syllabic boundaries results in an overrepresentation of word-initial and word-final syllables, while word-medial consonant clusters remain unobserved, leaving a substantial portion of the syllabic structures uttered in natural speech unaccounted for.

In the present study, we aim to overcome these two pitfalls by providing a quantitative estimate of the cross-linguistic prevalence of the Sonority Sequencing Principle (SSP) that takes into account the frequency with which speakers encounter SSP-following and SSP-violating structures in natural language. To this end, we considered a measure of syllabic token frequency, a measure of how often a particular syllable is uttered in natural speech in a given language per million words, which has been shown to be relevant in language processing and production tasks (Cholin & Levelt, 2009; Duchon et al., 2013; Perea & Carreiras, 1998). Additionally, to extract the syllabic repertoire of each language in the dataset, we used a large sample of syllabified

phonological transcriptions that include word-medial syllables, word-medial syllables which were not always represented in previous work. We then used a Bayesian modelling approach to estimate the prevalence of the SSP in a large dataset comprising 52,588 syllable types across 42 languages, yielding estimates of the proportion of SSP-following syllables in each language adjusted for the relative frequency of each syllable in natural speech.

Methods

Raw data, reproducible code, and project documentation are available in the Open Science Framework (https://osf.io/yeqms) and GitHub (https://github.com/NeuroDevCo/ssp-corpus).

Data collection and preprocessing

Phonological transcriptions were scraped from word entries in the Wiktionary database. Wiktionary is an open-source, free database containing 10,050,751 lexical entries from 8,244 languages contributed by 4,463,976 users (as of June 10th, 2026), many of which include at least one phonological transcriptions for a given word. The wikipron Python package (version 1.3.3, Lee et al., 2020) was used to scrape all available phonological transcriptions of words for all 7,859 languages included in the Glottolog database (Nordhoff & Hammarström, 2011). This resulted in approximately 1.7 million phonological transcriptions across 900 languages. Languages with less than 100 words were excluded, resulting in a final dataset of 731,145 words from 192 languages.

Available lexical frequencies for each word in each language were extracted using the *wordfreq* Python package (Speer, 2022), which provides an interface for retrieving lexical frequency data for words in over 40 languages. When lexical frequencies were not available for a particular language (e.g., the ISO 639-3 code did not match any language available in *wordfreq*), the closest equivalent language was

used, if available. Equivalent languages were identified based on their distance from the original language according to the CLDR v38.11 standard[1]. The rationale for using equivalent languages to retrieve missing lexical frequencies was that users of close languages are likely to use similar words to refer to the same objects, and lexical frequencies were therefore expected to be reasonably comparable across equivalent languages. Words with missing lexical frequency scores were excluded from the dataset.

The phonological transcriptions of words were then segmented into syllable candidates. When possible, syllable boundary markers were used, either explicit (“.”) or implicit (“ˈ” for primary stress, “ˌ” for secondary stress). In the absence of syllable markers, the word was assumed to be monosyllabic. Segments of each syllable candidate were then classified as consonants (C), vowels (V), or semi-vowels (v). Individual vowels, consecutive strings of vowels and semi-vowels (e.g., V, VV, Vv, VVv) were classified as vocalic clusters. Syllable candidates with more than one vocalic cluster (e.g., CVvCV, VCV) were flagged as multisyllabic and excluded from the dataset. This methods allowed to extract word-medial syllables when possible. Duplicated syllables within each language were removed (e.g., if the syllable /ka/ appeared in the Spanish words /ˈka.sa/ and /kaˈpo/, it was counted only once in the Spanish dataset). This procedure yielded one syllabic repertoire for each language.

To evaluate whether each extracted syllable followed or violated the SSP, a sonority score was assigned to each phoneme on a scale from one to nine, using the *panphon* Python package (version 0.22.2, Mortensen et al., 2016). Syllables with more

[1]The CLDR system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål → Norwegian, score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål → Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans → Dutch, Wu Chinese → Mandarin Chinese), etc.

than one peak (i.e., exhibiting a local minimum in the sonority contour) were classified as SSP-violating, and as SSP-following otherwise. Following this rule, syllables with a flat sonority contour (e.g., English word sphinxs /ˈsfɪŋks/ in which the first two phonemes bear identical sonority index value, 3-3-8-5-1-3) were classified as SSP-following. Whether consecutive segments with same sonority index comply or not with the SSP is a matter of debate (e.g., Tropf, 1987; Berent et al., 2007; Eckman, 1987; Santolin et al., 2023), and this reason we report complementary analyses in which flat sonority contours were classified as SSP violating, showing equivalent results (see Appendix C).

Table 1

Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically. For example, the syllable /'skul/ in Assamese shows a violation at onset, as sonority drops from /s/ (sonority index 3) to /k/ (sonority index 1) before rising again at nucleus /u/ (sonority index 8).

Language	Syllable	Original word	Sonority index	
SSP-following				
Faroese	/aɻ/	altar /ˈaɻ̩.taɹ/	9-6	
Finnish	/berk/	Berg /ˈberk/	2-9-6-1	
Kankanaey	/bal/	balla /balˈla/	2-9-6	
Moksha	/alʲ/	альняка /ˈalʲ.nʲa.kə/	9-6	
Sanskrit	/beː/	कुबेर /ku.beː.ɾɐ/	2-9	
SSP-violating				
Assamese	/skul/	স্কুল /skul/	3-1-8-6	
German	/amts/	Amtsträger /ˈamtsˌtʁeːgɐ/	9-5-1-3	
English	/biːdz/	beads /biːdz/	2-8-2-4	
Serbian-Croatian-Bosnian	/br̂ːzo/	brzo /br̂ːzo/	2-6-4-9	
Mon	/hma/	ဘျော /hma/	6-5-9	

To calculate the frequency of each syllable, we summed the lexical frequency (counts per million words) of all words in the dataset which contained that syllable (see Duchon et al., 2013 for a similar approach). The rationale behind this approach is that, whenever a word is uttered, all of its syllables are necessarily uttered as well. Therefore, summing the lexical frequencies of all words containing a given syllable yields an

estimate of that syllable's frequency in natural speech. The final dataset comprised 52,588 syllables from 42 languages from eight distinct language macro-families.

Data analysis

We used a multilevel Bayesian regression model to estimate the probability of each syllable in the dataset following (or violating) the SSP, considering the language it belonged to and its syllabic frequency. Bayesian models estimate a probability distribution for each parameter in the model. This distribution, named *posterior* distribution, is the result of combining prior knowledge about the parameter—implemented as a *prior* dsitribution that the researcher specifies based on domain knowledge—with the observed data via the Bayes rule. The posterior distribution of a parameter (e.g., of a specific coefficient in the regression model) provides inferential information about how likely such parameter is to be equal or close to zero. To estimate the posterior distribution of the model we used Hamiltonian Monte Carlo (HMC), an algorithm that explores the posterior distribution of the model in an iterative way, as implemented in *PyMC3* (version 6.2.0, Patil et al., 2010) using four chains of 2,000 iterations, 1,000 of which were warm-up iterations.

We modelled the response variable y_{ij} (1 if the syllable was SSP-following, 0 otherwise) using a Bernoulli distribution with probability parameter θ_{ij}. We specified a linear regression model for θ_{ij} which included fixed regression coefficients for the intercept (α) and the slope of syllabic frequency (β), and random effects u_j and w_j for each fixed coefficient, respectively. We specified weakly informative priors for all parameters in the model (see Equation 1 in Appendix A).

We report the median and 95% credible interval (CrI) of the posterior distribution of the model parameters of interest. The 95% CrI can be interpreted as being 95% likely to contain the true value of the parameter (Kruschke & Liddell, 2018). If this interval

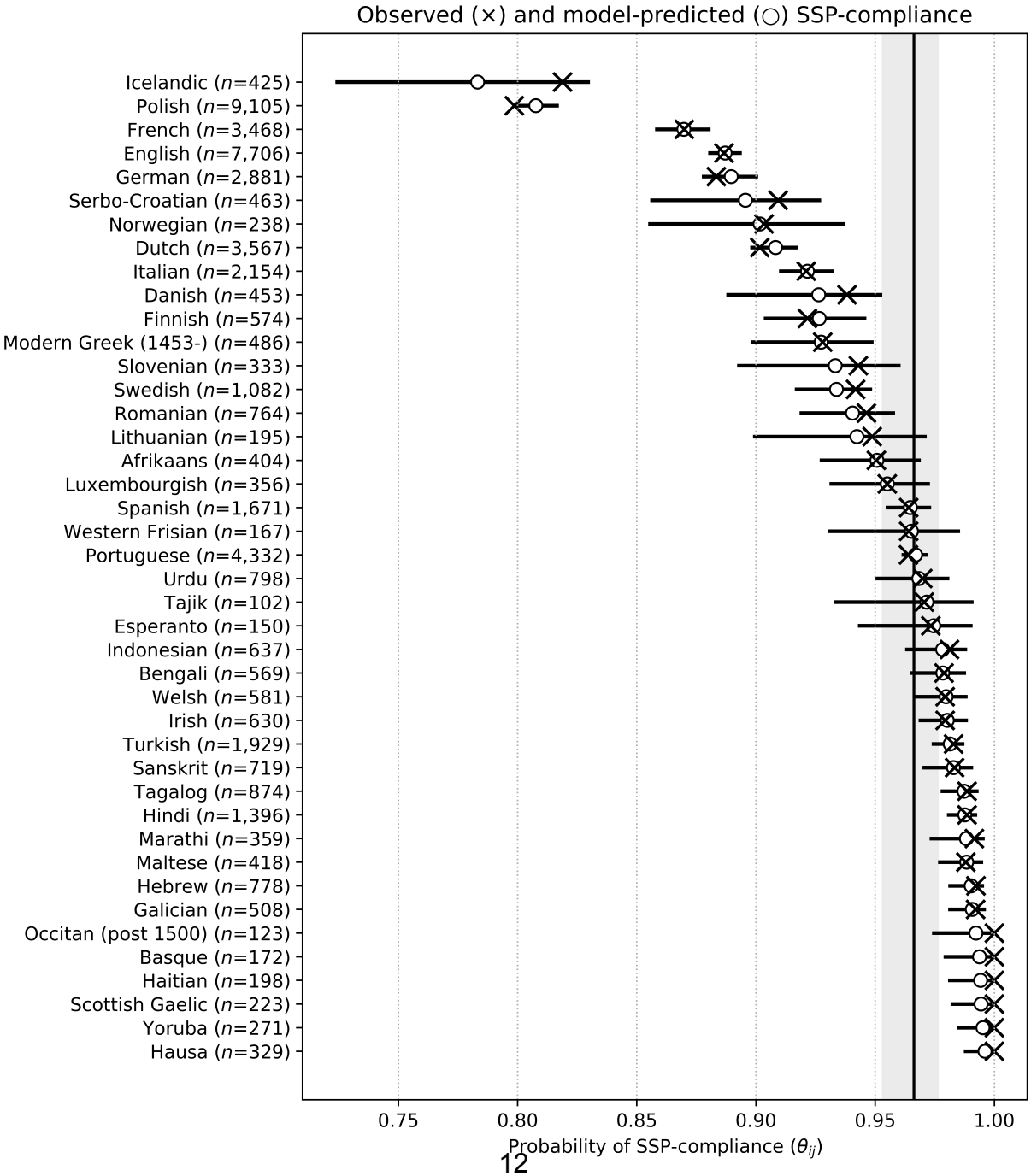
excluded zero, its corresponding parameter was concluded to be different from zero with 95% probability.

Results

The global proportion of SSP-following syllables across languages was θ = 0.966 (95% CrI = [0.953, 0.977]), and the prevalence of the SSP for each language ranged from 78.319% (Icelandic, 95% CrI = [72.353%, 81.730%]) to 99.605% (Hausa, 95% CrI = [98.720%, 99.918%]) (see Figure 2). Only 2.38% of the languages in the sample showed less than 80% of SSP prevalence, 14.29% languages showed less than 90% of SSP prevalence, and 38.10% languages showed less than 95% of SSP prevalence.

Figure 2

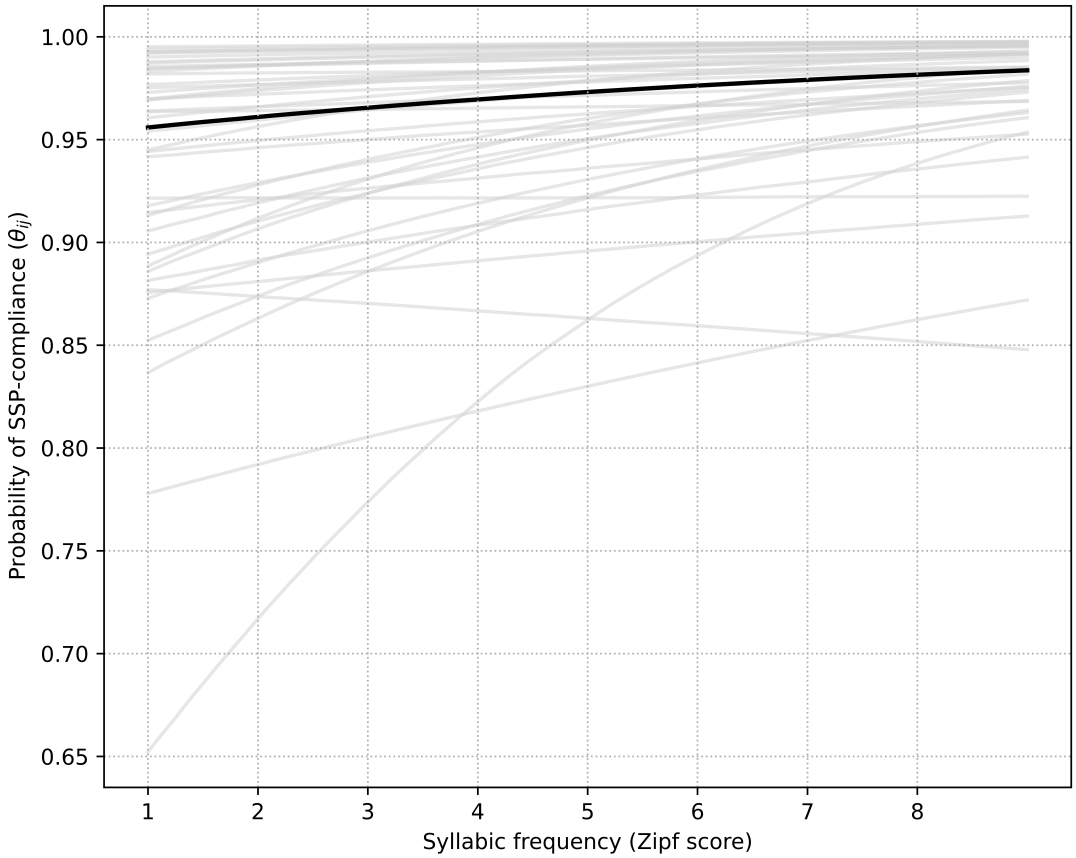
Observed (×) and model-predicted (○) estimates of SSP prevalence across languages in the dataset. Intervals indicate the 95% CrI of the posterior distribution. The number of syllables included in the dataset is indicated next to each language. The global estimate of SSP compliance and its 95% CrI are indicated as a vertical line and vertical rectangle.



Syllabic frequency showed a positive association with the probability of SSP compliance (β = 0.151, 95% CrI = [0.064, 0.246]), indicating that SSP-following syllables were, on average, more frequent than SSP-violating syllables (see Figure 3).

Figure 3

Mean predictions of SSP compliance as a function of syllabic frequency. Predicted global probability of SSP compliance is shown as a black line. Grey lines indicate language-specific predictions. Growing lines indicate a positive association between the frequency of a language's syllables and their probability of following the SSP.



Discussion

Previous experimental studies had pointed to the Sonority Sequencing Principle (SSP) as a universal constraint defining the structure of syllables, which are key units in speech processing (e.g., Berent & Lennertz, 2010; Berent et al., 2011; García-Castro et al., 2026; Gomez et al., 2014). However, it remained unclear to what extent SSP-following syllables are actually more common than SSP-violating ones across languages. The present study was aimed at quantifying the cross-linguistic prevalence of the SSP across a diverse set of languages using syllabified phonological transcriptions that include all syllables composing words (not just word-initial or final syllables) while also accounting for how often syllables are uttered in natural speech. Results revealed that SSP-following syllables are, on average, significantly more frequent than SSP-violating syllables across the languages in our dataset, with an overall 96.63% conformity with the SSP across languages. In addition, we found that syllables from most languages included in our sample followed the SSP more than 90% of the time.

A key novel aspect of the current study compared to prior work is the use of token frequencies instead of type frequencies. By using token frequencies, the estimates of SSP prevalence reported in this study more likely reflect how often speakers of a given language encounter SSP-following or SSP-violating syllables, compared to type frequencies. If SSP-following syllables occur more often in natural speech than SSP-violating syllables, estimates of SSP-prevalence likely underestimate their true value if token frequencies are not incorporated into the analysis. This may partially explain the higher prevalence of SSP compliance we found compared to other studies (e.g., 72% of SSP-following consonant clusters in 47 languages, Frisch, 2015; see also Watson, 2002; Al Tamimi & Al Shboul, 2013; Gishri, 2009; Yin et al., 2023). In

support of this possibility, in our analysis conducted with syllable token frequencies instead of type frequencies we found a positive, non-zero association between token syllabic frequency and the probability of SSP compliance.

Since SSP-following syllables are more frequent than SSP-violating ones across languages in our dataset, our results suggest that most languages comply with the SSP, albeit to varying degrees, and that violations of the SSP are exceptional for the majority of the languages included in our dataset. Complementarily, we also found that SSP violations, while uncommon, are attested in the majority of the languages in our dataset (see Yin et al., 2023 for a similar patter of results). We suggest that these results are best reconciled under the Optimality Theory (OT) framework (Frisch, 2015; Prince & Smolensky, 2004), according to which linguistic universals, rather than being treated as hard, binary rules, are instead conceptualized as a general bias towards certain type of linguistic structures that most languages favour. Exceptions to the SSP could therefore be understood as the result of the SSP interacting with other phonotactic rules that speakers of certain languages weight more strongly. Under OT, sonority contraints to syllablic structure remain useful to generate predictions about syllabic well-formedness despite the cross-linguistic variability found in the degree of compliance with this principle (Dell & Elmedlaoui, 1985).

A few points need to be considered when interpreting the findings in the present study. First, syllables included in our dataset were not randomly sampled. Instead, each syllable was included based on the availability of at least one word containing that syllable in its corresponding language in Wiktionary. Because of this, our dataset may be, to some extent, subject to selection bias (see Yin et al., 2023 for a similar argument). Second, while the present study includes multiple languages from a range of different language macro-families, the number of observations gathered for some

languages is reduced, especially when comparing Indo-European languages with languages from other macro-families. This unequal availability of data across languages is partially accounted for by the partial pooling of random effects that multilevel models implement (see Gelman et al., 2007 for a detailed explanation).

Overall, the present study updates prior estimates of cross-linguistic compliance with the SSP, pointing to both a substantially higher presence of SSP-following syllables than SSP-violating syllables across languages, and to the former syllables occurring more frequently in natural speech than the latter. It has been proposed that the learning mechanisms underlying language acquisition have shaped the structure of natural languages over time such that the organization of natural languages reflects the very abilities humans use to learn them (Chater & Christiansen, 2010; Christiansen & Chater, 2008, 2016; Saffran, 2001, 2002). Our findings are consistent with this view: the syllabic structures most prevalent across language families (SSP-following) are also those that human learners show a preference for from the earliest stages of language development.

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Appendix A

Model structure

Likelihood:

Global prevalence of the SSP:

$$
\begin{aligned}
y_{ij} &\sim \text{Bernoulli } (\frac{\theta_{ij}}{1-\theta_{ij}}) \\
\theta_{ij} &= \alpha + u_j + (\beta + w_j) \cdot \text{Frequency}_{ij}
\end{aligned}
$$

Prior:

$$
\begin{aligned}
\alpha, \beta &\sim \text{Normal } (0, 1) \\
u_j, w_j &\sim \text{Normal } (0, \sigma_{u,w}) \\
\sigma_{u,w} &\sim \text{HalfNormal } (0.5)
\end{aligned}
\tag{1}
$$

Where:

i indexes each observation and j indexes the language it belongs to

y_{ij} indexes whether the syllable follows (1) or violates (0) the SSP

u_j, w_j are language-specific adjustments to α and β

Appendix B

Number of SSP-following and SSP-violating syllables per language

Table B1

Number of SSP-following and SSP-violating syllables for each language in the dataset.

ISO-639-3	Words	Syllables	SSP+	%	ISO-639-3	Words	Syllables	SSP+	%
oci	243	123	123	100.0%	spa	106,961	1,671	1,611	96.4%
hat	848	198	198	100.0%	fry	425	167	161	96.4%
yor	1,311	271	271	100.0%	por	99,468	4,332	4,176	96.4%
hau	738	329	329	100.0%	ltz	860	356	340	95.5%
gla	1,112	223	223	100.0%	afr	619	404	384	95.0%
eus	10,555	172	172	100.0%	lit	279	195	185	94.9%
heb	948	778	772	99.2%	ron	2,126	764	723	94.6%
glg	3,477	508	504	99.2%	slv	1,704	333	314	94.3%
mar	703	359	356	99.2%	swe	1,624	1,082	1,019	94.2%
tgl	13,663	874	864	98.9%	dan	1,399	453	425	93.8%
hin	5,072	1,396	1,380	98.9%	ell	3,104	486	451	92.8%
mlt	3,777	418	413	98.8%	fin	73,343	574	529	92.2%
san	2,831	719	707	98.3%	ita	15,211	2,154	1,984	92.1%
tur	5,108	1,929	1,896	98.3%	hbs	8,822	463	421	90.9%
ind	3,851	637	625	98.1%	nor	451	238	215	90.3%
gle	4,682	630	617	97.9%	nld	11,340	3,567	3,216	90.2%
cym	5,150	581	569	97.9%	eng	36,929	7,706	6,832	88.7%
ben	2,771	569	557	97.9%	deu	13,142	2,881	2,545	88.3%
epo	5,975	150	146	97.3%	fra	17,659	3,468	3,017	87.0%
tgk	258	102	99	97.1%	isl	2,346	425	348	81.9%
urd	1,088	798	774	97.0%	pol	123,564	9,105	7,271	79.9%

Appendix C

Results after classifying flat sonority contours as SSP-violating

Figure C1

Observed (crosses) and model-predicted (circles) estimates of SSP prevalence across languages in the dataset. Intervals indicate the 95% and 79% CrI of the posterior distribution. The number of syllables included in the dataset is indicated next to each language.

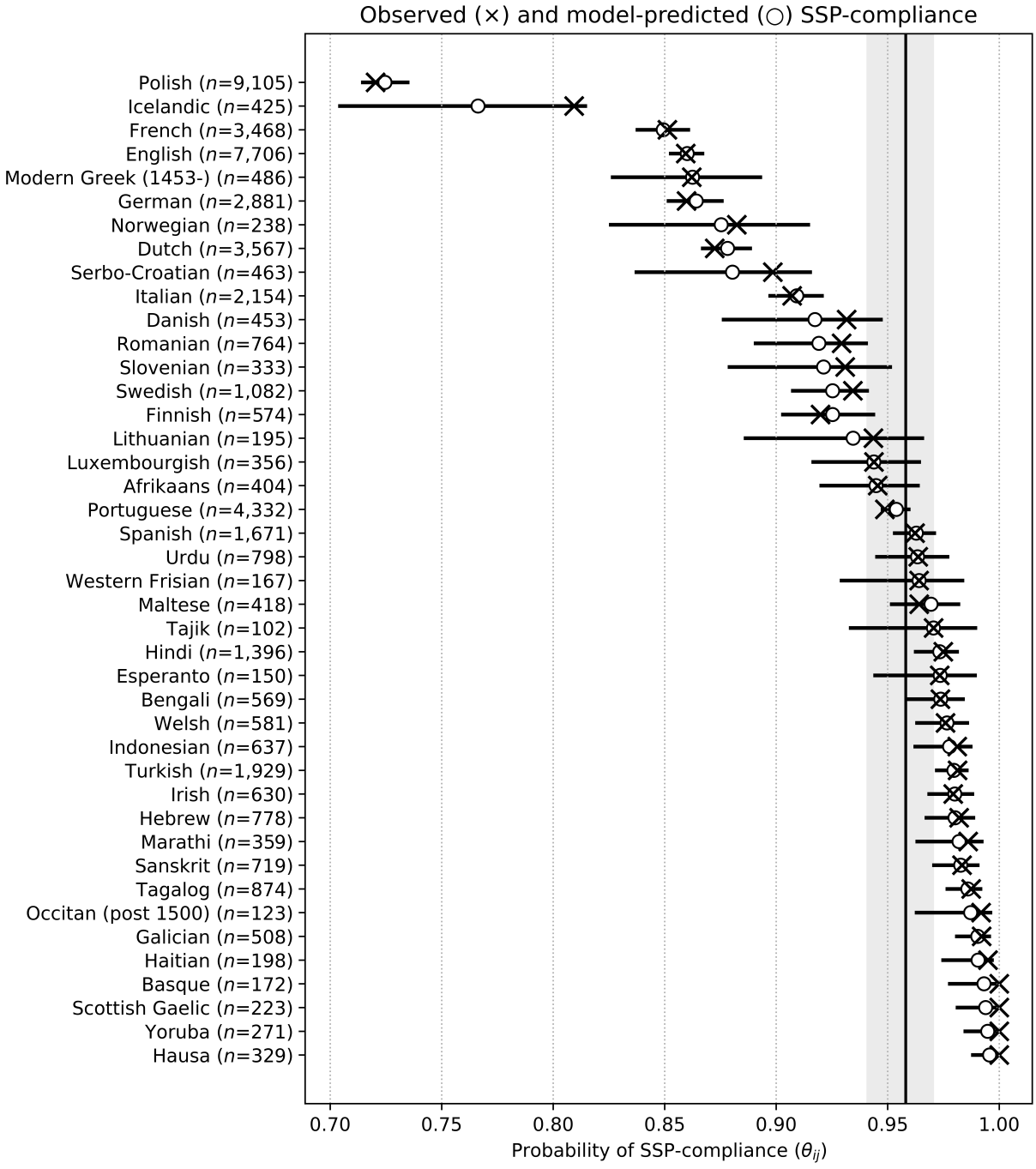


Figure C2

Mean predictions of SSP compliance as a function of syllabic frequency. Predicted cross-language probability of SSP compliance ($\theta_i j$) is shown as a black line. Thin lines indicate language-specific predictions, color-coded by language macro-family.

